

Condition Loops

Look at this code:

```
target = 7
attempts = 0
guess = 0
while guess != target and attempts < 3:
    guess = int(input("Guess 1 to 10: "))
    attempts = attempts + 1
print("Finished in", attempts, "tries.")
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Change the example so the final message explains why the loop stopped. After the loop, use an `if` statement to display "Correct!" when `guess` equals `target`; otherwise display "Out of guesses. It was 7." Test 7, then test 1, 2 and 3.

Challenge 2

Extend the example so each wrong guess gets a hint. Inside the `while` loop, use `if` and `elif` to display "Too low" or "Too high" after each guess that is not 7. For example, 5, 9 and 7 display "Too low", then "Too high".

Challenge 3

Create a door-code checker. Store "open" in `password`. Use a `while` loop to ask for a code until it is correct or three attempts are used. Display "Door opened." or "Door locked." followed by the attempts used. For example, "red", "blue" and "green" display "Door locked. Attempts used: 3".

